

Let's start by reviewing some properties of matrix operations:

I. $\bar{r}^T \bar{r} = \bar{r} \cdot \bar{r}$ dot product

$$\bar{r} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}, \quad \bar{r}^T \bar{r} = [r_1 \ r_2 \ \dots \ r_m] \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} = r_1^2 + r_2^2 + \dots + r_m^2$$

sum of squares $\rightarrow \sum_{i=1}^m r_i^2$

II. Transpose of Matrix sums

$$(A+B)^T = A^T + B^T$$

III. Distributivity

$$A(B+C) = AB + AC$$

IV. Transpose of ~~prose~~ product

$$(AB)^T = B^T A^T$$

V. Associativity of products

$$(AB)C = A(BC); ABC$$

VI

$A^T A$ is symmetric

Let $B = A^T A$

$$B^T = (A^T A)^T = \underbrace{(A^T)^T}_{A} (A^T)^T = A^T A$$

Apph IV

Linear regression fits the line, plane, or hyperplane to the data.

$$\hat{y}^{(i)} = h_{\theta}(\bar{x}^{(i)}) = \theta_0 + \theta_1 x_1^{(i)} + \dots + \theta_n x_n^{(i)} \quad (1)$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_n \end{bmatrix}, \quad \bar{x}^{(i)} = \begin{bmatrix} 1 \\ x_1^{(i)} \\ \vdots \\ x_n^{(i)} \end{bmatrix} \quad \text{then} \quad \hat{y}^{(i)} = \theta^T \bar{x}^{(i)}$$

To evaluate samples in parallel, we represent the samples as a matrix.

"design matrix" $\rightarrow X$

$$X = \begin{bmatrix} 1 & x_1^{(1)} & x_2^{(1)} & \dots & x_n^{(1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(m)} & x_2^{(m)} & \dots & x_n^{(m)} \end{bmatrix}$$

Each row is a sample in the training set.

It is convention to represent the samples as rows. If instead we had represented them as columns, let's call the transposed design matrix $Z = X^T$ then in matrix form (1) for multiple samples becomes

$$\hat{\bar{y}} = \theta^T Z \quad (1)$$

However with the row per sample convention (1) would become

$$\hat{\bar{y}} = X\theta \quad \text{where} \quad \hat{\bar{y}} = \begin{bmatrix} \hat{y}^{(1)} \\ \hat{y}^{(2)} \\ \vdots \\ \hat{y}^{(m)} \end{bmatrix} \quad \begin{matrix} \uparrow \\ m \text{ samples} \\ \downarrow \end{matrix} \quad (2)$$

Cost function $J(\theta)$

3

$$J(\bar{\theta}) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y)^2$$

(3)

using matrices, this is equivalent to

$$J(\bar{\theta}) = \frac{1}{2m} (\hat{\bar{y}} - \bar{y})^T (\hat{\bar{y}} - \bar{y})$$

↑
vector of predictions

(4)

~~Substitute (3) into (4) to get~~

$$\underline{J(\bar{\theta}) = \frac{1}{2m} (X\bar{\theta} - \bar{y})^T (X\bar{\theta} - \bar{y})}$$

Let's delay this substitution.

$$\nabla_{\theta} \left(\frac{1}{2m} (\hat{\bar{y}} - \bar{y})^T (\hat{\bar{y}} - \bar{y}) \right) = \frac{1}{2m} \nabla_{\theta} \underbrace{((\hat{\bar{y}} - \bar{y})^T (\hat{\bar{y}} - \bar{y}))}_{(5)}$$

(5)

Apply II to 5a

$$(\hat{\bar{y}}^T - \bar{y}^T) (\hat{\bar{y}} - \bar{y}) = (\hat{\bar{y}}^T - \bar{y}^T) (\hat{\bar{y}} - \bar{y})$$

(6)

Apply III to (6)

$$\hat{\bar{y}}^T \hat{\bar{y}} - \hat{\bar{y}}^T \bar{y} - \bar{y}^T \hat{\bar{y}} + \bar{y}^T \bar{y}$$

(7)

Reintroduce the nabla ∇ from (5)

$$\frac{1}{2m} \nabla_{\theta} \left(\underbrace{\hat{\bar{y}}^T \hat{\bar{y}}}_{(b)} - \underbrace{\hat{\bar{y}}^T \bar{y}}_{(c)} - \underbrace{\bar{y}^T \hat{\bar{y}}}_{(d)} + \bar{y}^T \bar{y} \right)$$

(8)

Note that we are taking the gradient with respect to the parameters Θ . ~~So~~ $\bar{y}$ is the ground truth. It doesn't change w/ Θ . Only $\hat{y}$ changes with Θ .

$\therefore$ (8) (d) becomes

$$\nabla_{\Theta}(y^T y) = 0$$

(8) (c) is $-\nabla_{\Theta}((\hat{y}^T y) + (\hat{y}^T \hat{y}))$

Apply IV to (8) (c) yields

$$-\nabla_{\Theta} \left[(\hat{y}^T y) + (\hat{y}^T \hat{y})^T \right] \quad (10)$$

Also scalar.

This looks like

a ~~scalar~~ dot product.

Dot product is a scalar

Transpose of a scalar is the same scalar

(10) becomes

$$-\nabla_{\Theta}(2 \hat{y}^T y)$$

(7) is now

$$\frac{1}{2m} \left[\nabla_{\Theta} \underbrace{\hat{y}^T \hat{y}}_{(a)} - 2 \nabla_{\Theta} \underbrace{\hat{y}^T y}_{(b)} \right] \quad (11)$$

Substitute $\hat{y} = X\theta$ into $1/(\cdot)$ to get

5

$$\nabla_{\theta} \hat{y}^T \hat{y} = \nabla_{\theta} (X\theta)^T (X\theta) \quad (12)$$

Apply IV transpose of product to (12) to get

$$\nabla_{\theta} \theta^T X^T X \theta \quad (13)$$

let $A = X^T X$

$$\nabla_{\bar{x}} \bar{x}^T A \bar{x} = (A + A^T) \bar{x}$$

This looks a bit like

Aside

Ex: consider 2×2 matrices.

$$\nabla_x \left([x_1 \ x_2] \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

$$\nabla_x \left([x_1 \ x_2] \begin{bmatrix} a_{11} x_1 + a_{12} x_2 \\ a_{21} x_1 + a_{22} x_2 \end{bmatrix} \right)$$

$$= \nabla_x \left(\underbrace{a_{11} x_1^2 + a_{12} x_1 x_2 + a_{21} x_1 x_2 + a_{22} x_2^2}_f \right)$$

$$= \nabla_x f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2a_{11} x_1 + (a_{12} + a_{21}) x_2 \\ (a_{12} + a_{21}) x_1 + 2a_{22} x_2 \end{bmatrix}$$

$$= x_1 \begin{bmatrix} 2a_{11} \\ a_{12} + a_{21} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} + a_{21} \\ 2a_{22} \end{bmatrix} = \begin{bmatrix} 2a_{11} & a_{12} + a_{21} \\ a_{12} + a_{21} & 2a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= (A^T + A) x = (A + A^T) x$$

$$\nabla_{\Theta} \Theta^T A \Theta = (A + A^T) \Theta = (X^T X + (X^T X)^T) \Theta = 2 X^T X \Theta \quad (14)$$

Now let's solve for (11)(b)

$$\nabla_{\Theta} y^T \hat{y} = \nabla_{\Theta} y^T X \Theta = \nabla_{\Theta} \underbrace{(y^T X)}_{\text{This is a row vector.}} \Theta \quad (15)$$

Let $\mathbf{a}^T = y^T X$

$$\nabla_{\Theta} \mathbf{a}^T \Theta = \nabla_{\Theta} [\alpha_0 \dots \alpha_n] \begin{bmatrix} \Theta_0 \\ \vdots \\ \Theta_n \end{bmatrix} = \nabla_{\Theta} (\alpha_0 \Theta_0 + \dots + \alpha_n \Theta_n)$$

$$= \nabla_{\Theta} f = \begin{bmatrix} \frac{\partial f}{\partial \Theta_0} \\ \vdots \\ \frac{\partial f}{\partial \Theta_n} \end{bmatrix} = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} = \mathbf{a}$$

So (15) becomes

$$\nabla_{\Theta} y^T \hat{y} = (y^T X)^T = X^T y \quad (16)$$

Substituting (16) & (14) into (11) yields

$$\frac{1}{2m} [2 X^T X \Theta - 2 \cancel{y^T X} X^T y] = 0$$

$$X^T X \Theta = X^T y$$

$$\underbrace{(X^T X)^{-1} (X^T X)}_I \Theta = (X^T X)^{-1} X^T y$$

$$\Theta = (X^T X)^{-1} X^T y$$

Normal Equation

The normal equation is a closed-form solution to find the optimal parameters for minimizing $J_{\Theta}(\Theta)$.

Time Complexity of the Normal Equation

$$\Theta = \underbrace{(X^T X)^{-1}}_{T_1} X^T y$$

T_1 = matrix multiplication of X^T and X

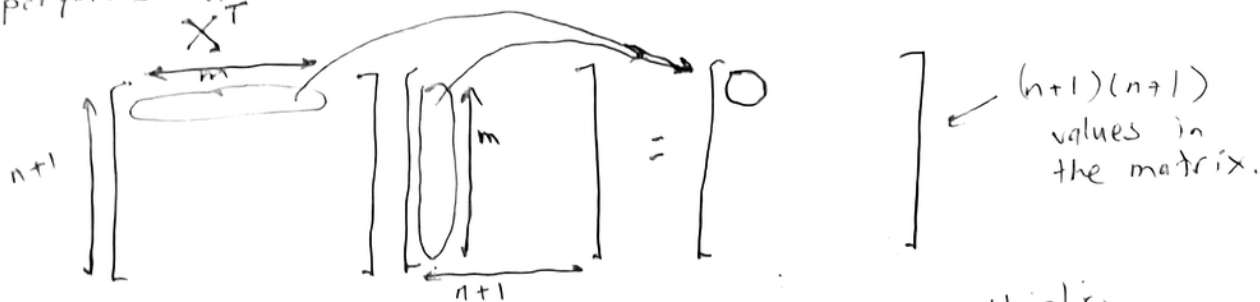
T_2 = inverting the matrix resulting from step 1.

T_3 = multiplying the inverted matrix resulting from step 2.

T_4 = multiplying the matrix from step 3 by y

$$T = T_1 + T_2 + T_3 + T_4$$

A straight forward implementation of matrix multiplication performs a dot product of ^{each} every row with each column.



A single dot product takes ~~one~~ m multiplies and ~~one~~ $m-1$ additions. A single dot product of a column with a row is $O(m)$

However ~~these~~ are the resulting matrix is
 $(n+1) \times (n+1)$ so there are $n^2 + 2n + 1$ values.

$$T_1 = O(m \cdot (n+1)^2) = O(mn^2).$$

For multiplying to 2 square matrices this is $O(n^3)$.
~~The best known~~ with Strassen's algorithm this can be
 reduced to $O(n^{2.8074})$. This is still expensive.

T_2 ~~requires~~ requires performing matrix inversion.
 On an $n \times n$ matrix using Gaussian elimination or LU
 factorization, matrix inversion is $O(n^3)$.

The matrix output from step 1 has dimensions
 $(n+1) \times (n+1)$ which is still $O(n)$ in each dimension so

$$T_2 = O(n^3)$$

There are ways to avoid the inversion, but they are
 still $O(n^3)$

$$T_3 \Rightarrow \begin{matrix} \xrightarrow{n+1} \\ \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \end{matrix} \cdot \begin{matrix} \xleftarrow{m} \\ \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \end{matrix} = O((n+1)(n+1)(m))$$

$$= O(n^2 m)$$

Matrix from step 3 has $(n+1) \times m$ dimensions

$$T_4 \Rightarrow \begin{matrix} \xleftarrow{m} \\ \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \end{matrix} \cdot \begin{matrix} \xrightarrow{m} \\ \left[\begin{array}{c} \vdots \\ \vdots \end{array} \right] \end{matrix}$$

(m) per row-column dot
 product.
 there are $n+1$ dot products
 so
 $O((n+1)m) = O(nm)$

$$T = T_1 + T_2 + T_3 + T_4$$

$$T = O(mn^2) + O(n^3) + O(n^2m) + O(nm)$$

$$T = O(mn^2 + n^3 + n^2m) \quad \text{where } m \geq n$$

$$\boxed{T = O(mn^2 + n^3) \quad \text{where } m \geq n.}$$

Problems w/ Linear Regression

- 1) Only works well for linear relationships.
- 2) High computational cost to solve normal equation for large number of samples m .
- 3) High computational cost to solve normal equation for large number of dimensions n .

We can reduce m by using fewer samples or by ~~at~~ performing piecewise approximations (i.e., fit locally), but n is still a problem. When dimensionality is high, linear regression is maybe not appropriate.

When dimensions are not as meaningful (e.g., not orthogonal), n can be reduced with dimensionality reduction (outside the scope of this lecture).